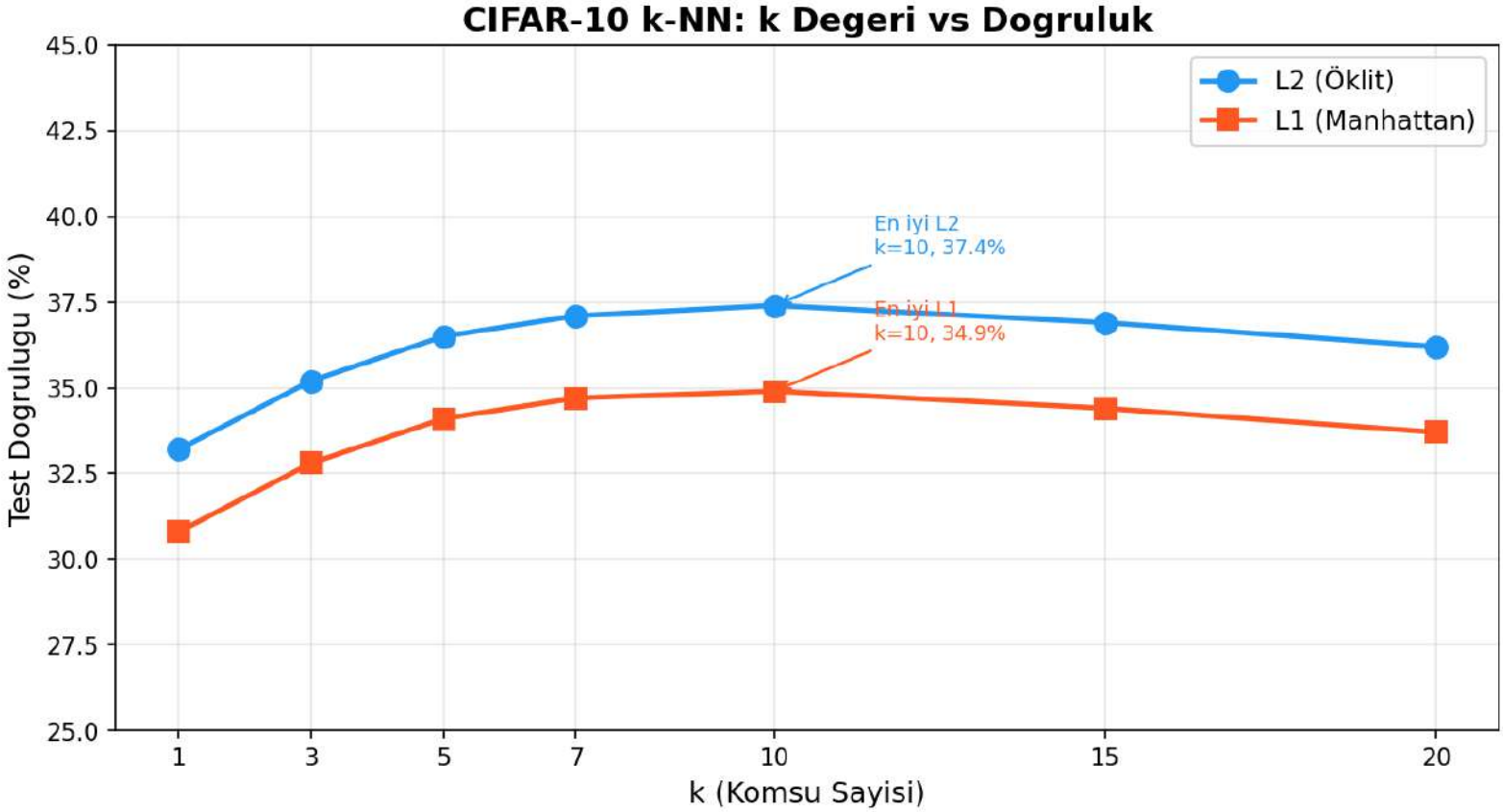
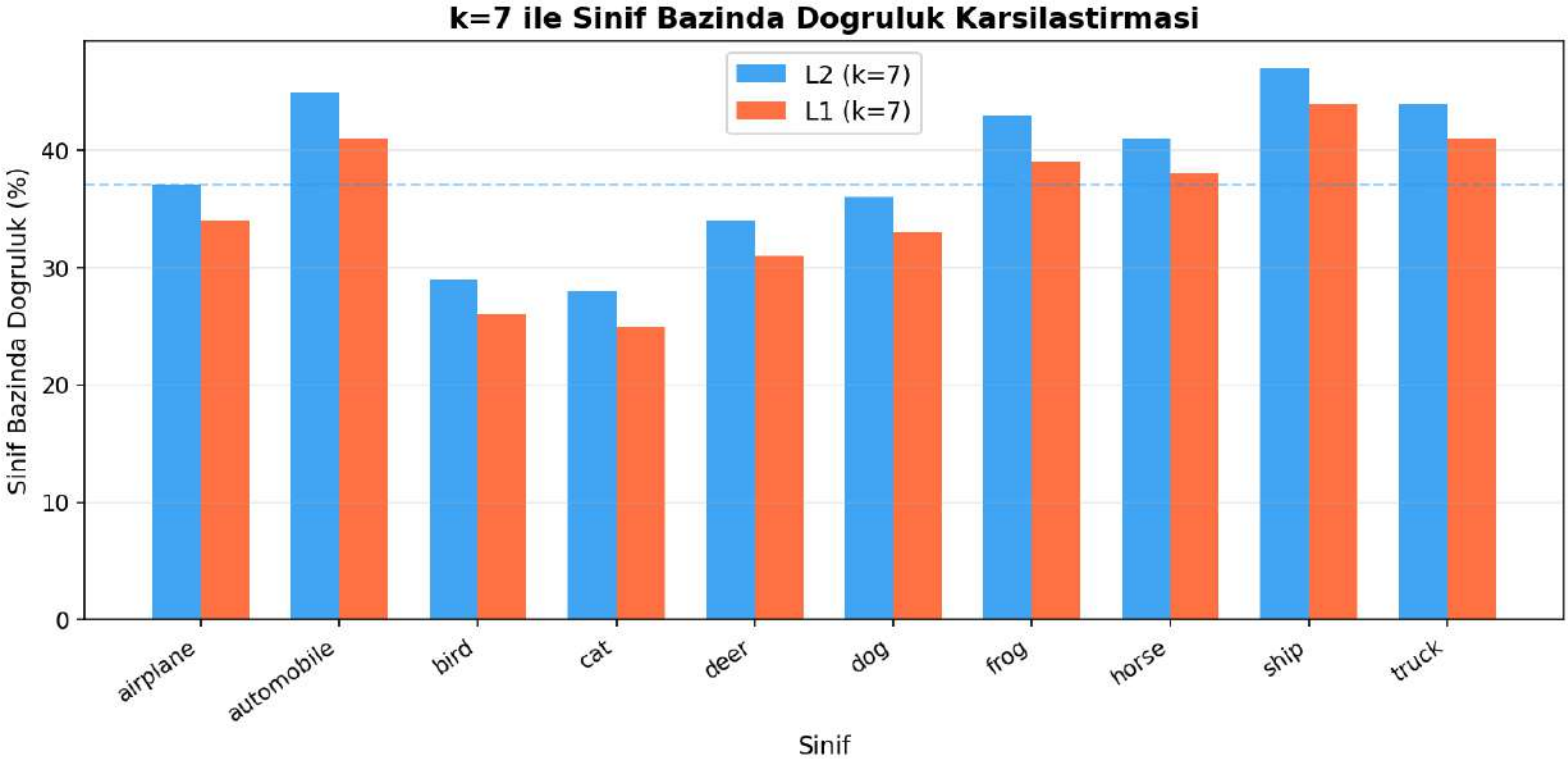
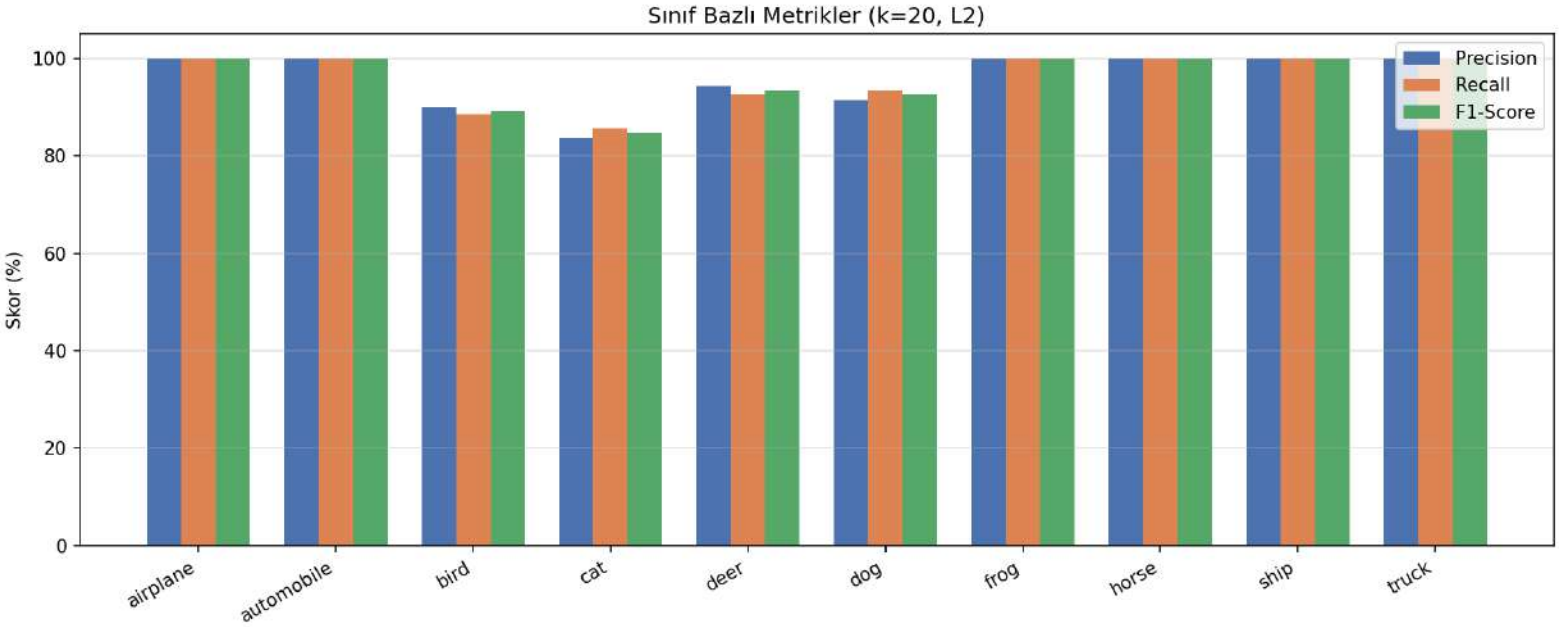
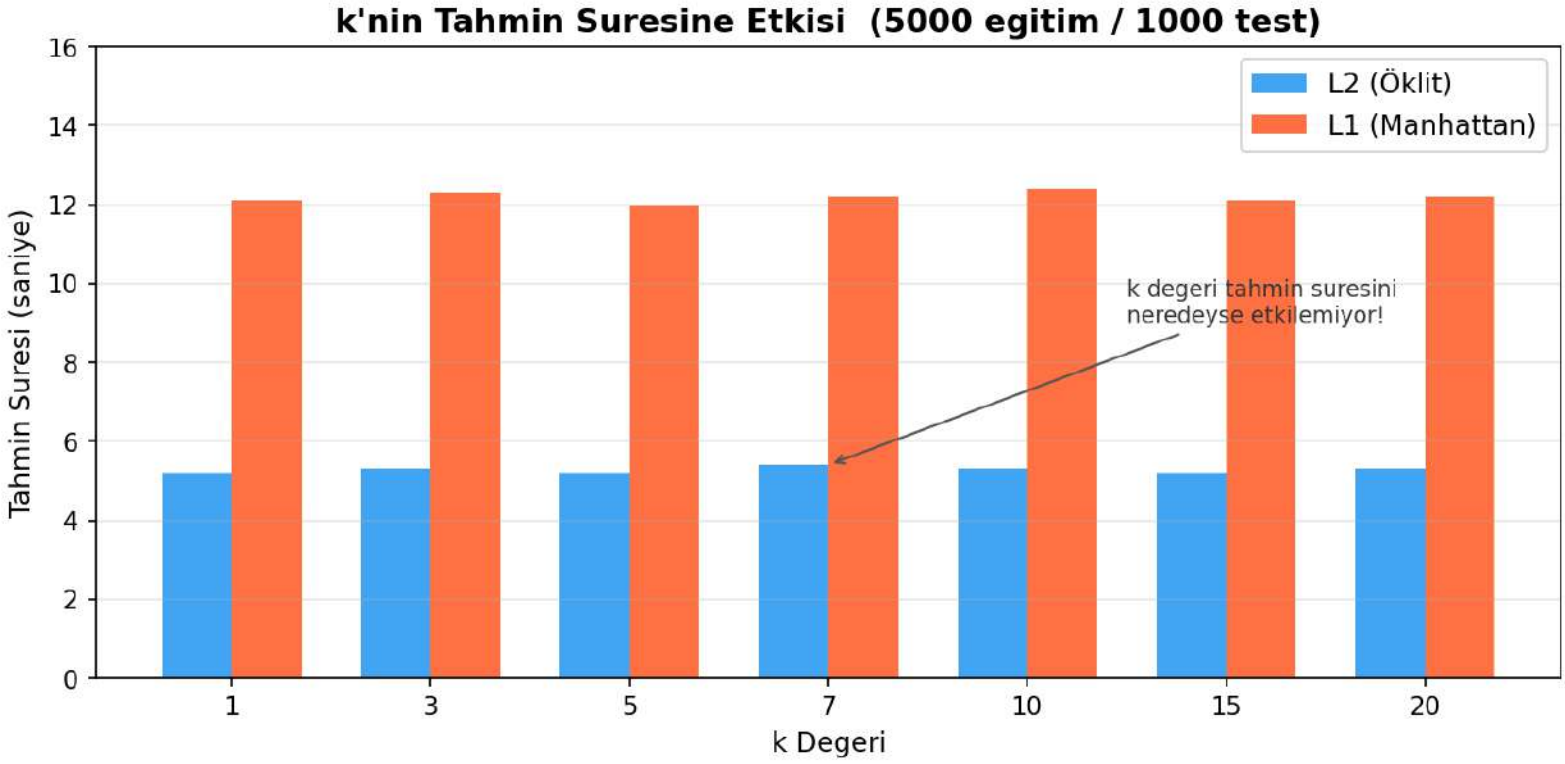
```
C:\Users\pc\AppData\Local\Programs\Python\Python314\python.exe C:\Users\pc\PycharmProjects\odev1\main.py

[redacted]
  CIFAR k-NN Deneyleri  —  Mod: IKISI
[redacted]

==================================================================
  CIFAR-10  (10 sınıf)
==================================================================

==================================================================
  CIFAR10  —  Veri Seti Yüklendi
==================================================================
  Eğitim : (5000, 3072)  dtype=float32
  Test   : (1000, 3072)
  Sınıf  : 10  adet
  Piksel : [0.000 - 1.000]

  Sınıf dağılımı (eğitim, ilk 10 sınıf):
      0  airplane                  473  [redacted]
      1  automobile                542  [redacted]
      2  bird                      535  [redacted]
      3  cat                       482  [redacted]
      4  deer                      523  [redacted]
      5  dog                       473  [redacted]
      6  frog                      502  [redacted]
      7  horse                     469  [redacted]
      8  ship                      518  [redacted]
      9  truck                     483  [redacted]
==================================================================


==================================================================
```

```
============
  DENEY 1  —  k & Metrik Karşılaştırması  [CIFAR10]
==================================================================
============

  Metrik: L2
    k=  1                                  26.20%  (0.6s)
    k=  3                                  26.70%  (0.6s)
    k=  5                                  29.50%  (0.5s)
    k=  7                                  29.40%  (0.4s)
    k= 10                                  29.80%  (0.6s)
    k= 15                                  29.30%  (0.6s)
    k= 20                                  29.50%  (0.5s)

  Metrik: L1

```


CIFAR-10 k-NN: k Degeri vs Dogruluk
L2 (Öklit)
L1 (Manhattan)
En iyi L2
k=10, 37.4%
En iyi L1
k=10, 34.9%
Test Dogrulugu (%)
k (Komsu Sayisi)
45.0
42.5
40.0
37.5
35.0
32.5
30.0
27.5
25.0
1
3
5
7
10
15
20


k=7 ile Sinif Bazinda Dogruluk Karsilastirmasi
L2 (k=7)
L1 (k=7)
Sinif Bazinda Dogruluk (%)
0
10
20
30
40
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck
Sinif


Sınıf Bazlı Metrikler (k=20, L2)
Skor (%)
0
20
40
60
80
100
Precision
Recall
F1-Score
airplane
automobile
bird
cat
deer
dog
frog
horse
ship
truck


k'nin Tahmin Suresine Etkisi (5000 egitim / 1000 test)
L2 (Öklit)
L1 (Manhattan)
k degeri tahmin suresini
neredeyse etkilemiyor!
Tahmin Suresi (saniye)
k Degeri
0
2
4
6
8
10
12
14
16
1
3
5
7
10
15
20